SYSTEM AND METHOD FOR N-GRAM PARALLEL DATA ENCODING

Reinaldo Javier Aguiar Marcano

March 9, 2026

Title: SYSTEM AND METHOD FOR N-GRAM PARALLEL DATA ENCODING

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Abstract

A system and method for encoding data on a high cardinality parallel communication channel by creating logical groupings of physical channels, termed "n-grams." The method comprises: defining a group of 'n' adjacent physical channels on a parallel interface; treating the combined state space of the 'n' channels as a single logical channel with a vastly expanded number of states; and mapping a large data block to a single state of this logical n-gram channel. For example, by treating two adjacent 24-bit pixels ("bigram") as a single logical channel, a 48-bit (6-byte) data block can be encoded in a single transmission step. This method provides an exponential increase in data density and enables a form of over-the-channel data compression.

Background of the Invention

High-cardinality parallel interfaces allow for the transmission of multi-byte data blocks in a single state of a single physical channel (e.g., one pixel). While this achieves a high level of data density, there is a need to encode even larger data blocks that may exceed the state-space of a single channel. For instance, a cryptographic key or a 64-bit data type is larger

than the 24 bits of data that can be encoded in a standard pixel. A method is needed to structure data across multiple parallel channels in a way that further increases the effective data density of the physical layer.

Summary of the Invention

The present invention is a system and method for n-gram parallel data encoding, which increases the data density of a parallel communication interface by creating logical channels from groups of physical channels. An "n-gram" is a defined grouping of 'n' adjacent physical channels (e.g., pixels) that are treated by the encoding system as a single, unified logical channel.

The method involves defining the size of the n-gram (e.g., n=2 for a "bigram," n=3 for a "trigram"). The total number of unique states that this logical n-gram channel can represent is the product of the states of its constituent physical channels. For example, if a single pixel can represent N states, a bigram of two pixels can represent N*N = N$\hat{2}$ unique states. A communication system can then map a data block of a correspondingly larger size to a single state of the logical n-gram. This allows for the encoding of, for instance, a 6-byte (48-bit) data block into the combined state of two adjacent 24-bit pixels. This technique achieves a form of data compression directly on the physical layer by maximizing the utilization of the parallel channel architecture.

~~Statement of License Grant to the United States Government~~

~~The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.~~

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the difference between single-channel encoding and n-gram encoding. **FIG. 2** is a block diagram showing the data flow for an n-gram

encoding system.

Detailed Description of the Invention

Referring to **FIG. 1**, a conceptual comparison is shown. In a standard single-channel encoding scheme (100), a first data block (101) is mapped to the state of a first physical channel (103), and a second data block (102) is mapped to the state of an adjacent second physical channel (104). In the n-gram encoding method (110) of the present invention, a single, larger data block (111) is mapped to the combined state of a logical group (112) comprising both the first physical channel (103) and the second physical channel (104). This logical grouping is an n-gram, in this case a bigram.

FIG. 2 illustrates the data flow. A stream of data (201) is received by an n-gram encoder (202). The encoder groups the data into large data blocks (203) suitable for n-gram encoding. For each block, the encoder determines the required combined state for a group of 'n' physical channels (204) and instructs the transmitter (205) to set those channels to the appropriate states. On the receiving end, the sensor (206) reads the states of the group of channels, and an n-gram decoder (207) uses the combined state to reconstruct the original large data block (208).

~~Claims~~

~~What is claimed is:~~

1. ~~A method for encoding data on a parallel communication interface having a plurality of physical channels, the method comprising the steps of:~~
    - ~~defining a logical channel as a group of 'n' physical channels from the plurality of physical channels, where n is greater than 1;~~
    - ~~determining a combined state-space of the logical channel based on the possible states of the 'n' constituent physical channels; and~~
    - ~~encoding a single data block by mapping it to a single unique state within the combined state space of the logical channel.~~
2. ~~The method of claim 1, wherein the physical channels are pixels in a two-dimensional array.~~

3. ~~The method of claim 2, wherein n=2, defining a "bigram" logical channel.~~

4. ~~The method of claim 3, wherein each pixel is capable of representing $2\hat{2}4$ states, and the bigram logical channel is capable of representing $2\hat{4}8$ states.~~

5. ~~The method of claim 4, wherein the single data block is a 6-byte data block.~~

6. ~~The method of claim 1, wherein the parallel communication interface is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.~~

Drawings



Figure 1: Conceptual diagram illustrating the difference between single-channel encoding and n-gram encoding.

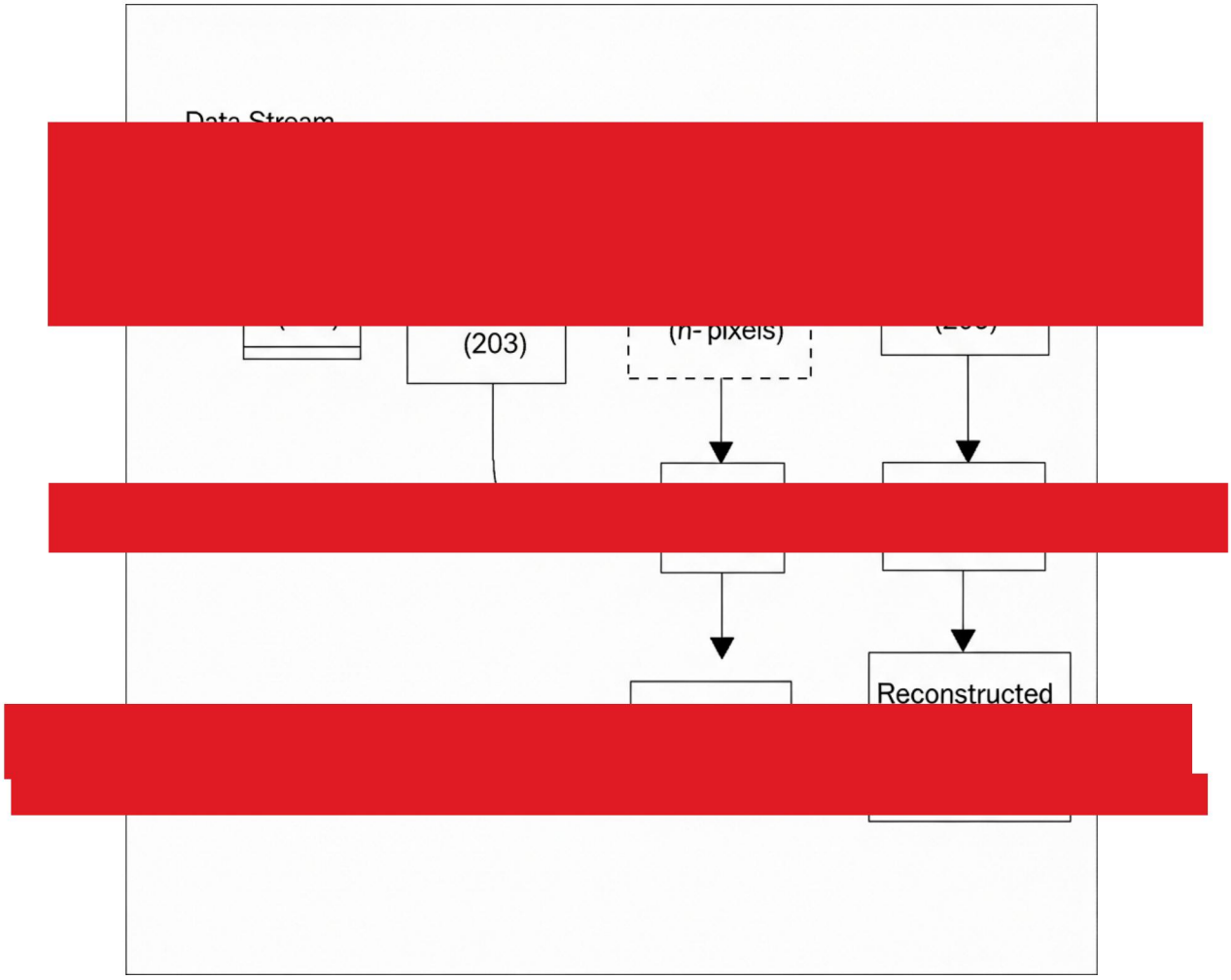


Figure 2: Block diagram showing the data flow for an n-gram encoding system.